

HW9 Solutions

ECE 310: Digital Signals Processing, Spring 2025

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Prepared by: Ben & Shilan

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Solution:

One can observe that y is obtained by (circularly) shifting x to the right by 3. In other words, $y[n] = x[\langle n - 3 \rangle_6]$. Then by the shifting property of DFT we have

$$Y[k] = X[k]e^{-j\frac{\pi}{3}k \cdot 3} = X[k]e^{-j\frac{2\pi}{6}k \cdot 3}$$

Therefore, a clear choice of n_0 is $n_0 = 3$. To obtain another valid n_0 , by using the periodicity, we have

$$Y[k] = X[k]e^{-j\frac{2\pi}{6}k \cdot 3}$$

which leads to $n_0 = 9$. In general, one can see that any n_0 that satisfies

$$n_0 = 3 + 6m, \quad m \in \mathbb{Z}$$

is valid (try to understand this from the perspective of timeshift).

Grading: 15 points

- +9 circular shift property
- +3 each correct n

2

Solution:

According to the circular time reversal property, $x[\langle -n \rangle_N] \longleftrightarrow X[\langle -k \rangle_N]$,

we can get $y[n] = x[\langle -n \rangle_6] = [1, 6, 5, 4, 3, 2]$.

Grading: 15 points

- +10 for identifying the correct property or manually deriving the reasonable relationship
- +5 for correct answer (+2 if obtain $[6, 5, 4, 3, 2, 1]$)

3

Solution:

We can call the circular convolution result of $x[n]$ and $y[n]$ as

$$\begin{aligned} z[n] &= x[n] \circledast y[n] \\ &= \sum_{m=0}^{N-1} x[m] y[\langle n - m \rangle_N] \\ &= \sum_{m=0}^{N-1} y[m] x[\langle n - m \rangle_N], \end{aligned}$$

and the inverse DFT of $X[k]Y[k]$ as $z'[n]$, such that

$$\begin{aligned} z'[n] &= \frac{1}{N} \sum_{k=0}^{N-1} Z'[k] e^{j \frac{2\pi k}{N} n} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] Y[k] e^{j \frac{2\pi k}{N} n} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} \left(\sum_{m=0}^{N-1} x[m] e^{-j \frac{2\pi k}{N} m} \right) \left(\sum_{l=0}^{N-1} y[l] e^{-j \frac{2\pi k}{N} l} \right) e^{j \frac{2\pi k}{N} n} \\ &= \sum_{m=0}^{N-1} \sum_{l=0}^{N-1} x[m] y[l] \left(\frac{1}{N} \sum_{k=0}^{N-1} e^{j(2\pi k) \frac{n-m-l}{N}} \right) \end{aligned}$$

The last line of previous equation can be further simplified by the below property

$$\frac{1}{N} \sum_{k=0}^{N-1} e^{j(2\pi k) \frac{n-m-l}{N}} = \begin{cases} 1, & \langle n - l - m \rangle_N = 0 \\ 0, & \text{otherwise} \end{cases}$$

Thus, we constrain that $l = \langle n - m \rangle_N$ or $m = \langle n - l \rangle_N$. By substituting $l = \langle n - m \rangle_N$, we can get

$$z'[n] = \sum_{m=0}^{N-1} x[m] y[\langle n - m \rangle_N] = z[n].$$

Grading: 15 points

- +5 for correctly applying the forward/backward DFT transform
- +5 for identifying the special property in the answer
- +5 for reasonably finishing the proof with the property
- Other valid proofs are acceptable.

4

Solution:

(a) To convert the signal to discrete-time, we use $x[n] = x_c(nT) = \cos(\frac{\pi}{5}n)$. However, note that we are only sampling for five seconds, so the signal is truncated and finite: $\{x[n]\}_{n=0}^{299}$. If $x[n]$ was not truncated, the DTFT of a cosine would be a pair of shifted delta functions for every period. Since there is truncation, the DTFT now has the deltas replaced with sinc-like functions.

Given the resemblance between the DTFT of the cosine and truncated cosine, notice that the greatest magnitudes for both DTFTs are at the same values of ω :

$$\frac{\pi}{5} + 2\pi m, \quad -\frac{\pi}{5} + 2\pi m \quad \forall m \in \mathbb{Z}$$

For $X[k]$, values of $0 \leq k < N$ map to values of $0 \leq \omega < 2\pi$. So, the values of ω between 0 and 2π with the greatest magnitude are $\pi/5$ and $9\pi/5$. Using the fact that $\omega = \frac{2\pi k}{N}$, we can find the integer values of k where $X[k]$ has the greatest magnitude:

$$\begin{aligned} \frac{\pi}{5} &= \frac{2\pi k}{300} \rightarrow k = 30 \\ \frac{9\pi}{5} &= \frac{2\pi k}{300} \rightarrow k = 270 \\ k &= 30, 270 \end{aligned}$$

(b) When $N = 512$, the solved values of k are no longer integers. Round to the nearest integer.

$$\begin{aligned} \frac{\pi}{5} &= \frac{2\pi k}{512} \rightarrow k = 51 \approx 51.2 \\ \frac{9\pi}{5} &= \frac{2\pi k}{512} \rightarrow k = 461 \approx 460.8 \\ k &= 51, 461 \end{aligned}$$

Grading: 15 points

- part (a) (10 points)
 - +4 for finding the correct greatest value of DTFT
 - +3 for each correct value of k
- part (b) (5 points)
 - +1.5 each if k values are not an integer or off by one
 - +2.5 for each correct value,

5

Solution:

(a) Each pair of peaks in the graph correspond to a cosine. Assume that the pair $k = 4$ and $k = 28$ are the DFT of $A_0 \cos(\Omega_0 t)$ and the pair $k = 12$ and $k = 20$ are the DFT of $A_1 \cos(\Omega_1 t)$. To find the frequencies of the continuous-time signal, recall that $\omega = \frac{2\pi k}{N}$ and $\Omega = \frac{\omega}{T}$. So:

$$\begin{aligned} \Omega_0 &= \frac{2\pi k}{NT} = \frac{2\pi 4}{32T} = 8\pi \times 10^{12} \\ \Omega_1 &= \frac{2\pi k}{NT} = \frac{2\pi 12}{32T} = 24\pi \times 10^{12} \end{aligned}$$

where $N = 32$ and $\frac{1}{T} = 32$ Terahertz. To find the amplitudes A_0 and A_1 , we first take the DFT of one discrete cosine function $x_0[n] = A_0 \cos(\omega_0 n)$.

$$X_0[k] = A_0 \frac{\sin\left(\frac{N}{2}\left(\frac{2\pi k}{N} - \omega_0\right)\right)}{2 \sin\left(\frac{1}{2}\left(\frac{2\pi k}{N} - \omega_0\right)\right)} e^{-j\frac{N-1}{2}\left(\frac{2\pi k}{N} - \omega_0\right)} + A_0 \frac{\sin\left(\frac{N}{2}\left(\frac{2\pi k}{N} + \omega_0\right)\right)}{2 \sin\left(\frac{1}{2}\left(\frac{2\pi k}{N} + \omega_0\right)\right)} e^{-j\frac{N-1}{2}\left(\frac{2\pi k}{N} + \omega_0\right)}$$

And find its magnitude:

$$|X_0[k]| = \left| A_0 \frac{\sin\left(\frac{N}{2}\left(\frac{2\pi k}{N} - \omega_0\right)\right)}{2 \sin\left(\frac{1}{2}\left(\frac{2\pi k}{N} - \omega_0\right)\right)} \right| + \left| A_0 \frac{\sin\left(\frac{N}{2}\left(\frac{2\pi k}{N} + \omega_0\right)\right)}{2 \sin\left(\frac{1}{2}\left(\frac{2\pi k}{N} + \omega_0\right)\right)} \right|$$

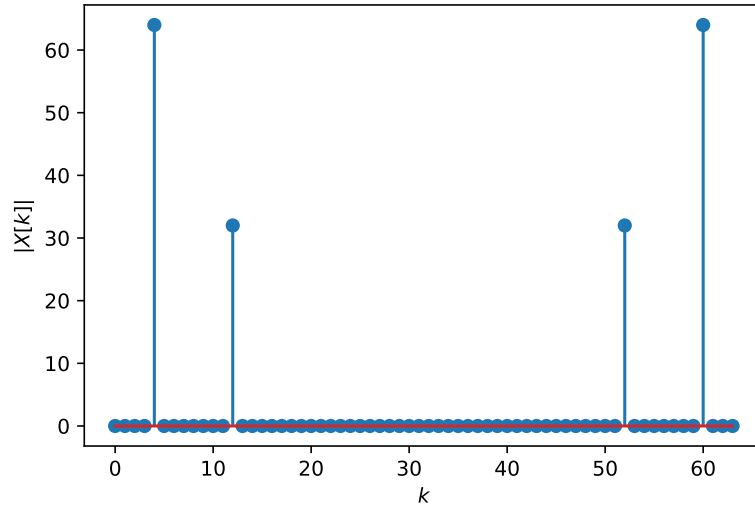
where $\omega_0 = \frac{\pi}{4}$. We know by the graph that there are peaks for $|X_0[k]|$ at $|X_0[4]| = |X_0[28]| = 32$. Using L'Hopital's rule, we find $|X_0[4]| = |X_0[28]| = \frac{A_0 N}{2}$. So, $A_0 = 2$. Repeating these steps to solve for A_1 results in $A_1 = 1$.

$$\Omega_0 = 8\pi \times 10^{12} \quad \Omega_1 = 24\pi \times 10^{12} \quad A_0 = 2 \quad A_1 = 1$$

(b) We now have a sampling rate of $\frac{1}{T} = 64$ Terahertz and $N = 64$. Using the answers from the last part, we can find each peak's new location and magnitude.

$$\begin{aligned} k_0 &= \frac{\Omega_0 N T}{2\pi} = 4 \\ k_1 &= \frac{\Omega_1 N T}{2\pi} = 12 \\ |X[k_0]| &= \frac{A_0 N}{2} = 64 \\ |X[k_1]| &= \frac{A_1 N}{2} = 32 \end{aligned}$$

Note that the spectral copies of the DTFT in the interval $(\pi, 2\pi)$ will also be present in the DFT. They are at $N - k_0 = 60$ and $N - k_1 = 52$. The plot of the new DFT magnitude is provided in figure below.



Grading: 25 points

- part (a) (8 points)
 - +2 each for correct answers of $\Omega_0, \Omega_1, A_0, A_1$ in (a)
 - +1 if minor math errors of each value
- part (b) (17 points)
 - +3 each for correct answers of $k_0, k_1, |X[k_0]|, |X[k_1]|$ in (b)
 - +1 if minor math errors in each value
 - +5 for a correct plot of the new DFT
 - +2.5 for minor error in plot of new DFT

Solution:

For index k and length N , we have the DFT evaluated the DTFT at frequency $k = \frac{2\pi k}{N}$. We may also relate digital frequencies to analog frequencies via $\omega = \Omega T$. For the provided values, we may then find

$$\Omega_1 = \frac{\omega_1}{T} = \frac{2\pi \cdot 8}{64 \cdot \frac{1}{8}} = 2\pi \text{rad/s}$$

$$\Omega_2 = \frac{\omega_2}{T} = \frac{2\pi \cdot 16}{64 \cdot \frac{1}{8}} = 4\pi \text{rad/s}$$

Next, each spectral component is represented by a periodic sinc function with peak value $\frac{AN}{2}$ for amplitude A and length N . Thus, we know that the respective heights of each spectral component in the DFT plot will be multiples of $\frac{N}{2} = 32$ since A_1 and A_2 are integers. Observing the plot, we may conclude that

$$A_1 = 1, A_2 = 3.$$

Grading: 15 points

- +3 for identifying the relationships $k = \frac{2\pi k}{N}$ and $\Omega = \frac{\omega}{T}$
- +3 for correct answer of A_1 and A_2 (+6 in total)
- +3 for correct answer of Ω_1 and Ω_2 (+6 in total)